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Riveted Composite Structures Under Static and Impact Loading: A Comparative Finite Element Study

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Abstract

Aircraft structures widely use Glass Fibre Reinforced Polymer (GFRP) facesheets combined with Nomex honeycomb core sandwich panels because of their high stiffness to weight efficiency. However, these structures are vulnerable to interlaminar delamination under quasi static loading and facesheet core debonding under transverse impact. This study numerically investigates the effectiveness of mechanical riveting as a damage mitigation strategy using finite element analysis.

Two complementary numerical approaches are employed. Quasi static delamination behaviour (Case 1) is analysed using ANSYS Composite PrepPost for a 12 ply quasi isotropic GFRP laminate, with a bilinear cohesive zone model introduced at the laminate mid plane. Three rivet configurations, namely single rivet, two rivet and four rivet layouts, are evaluated under both tensile load controlled and displacement controlled loading conditions, with a maximum applied tensile load of 20 kN. The results show that increasing rivet count improves load redistribution and global stiffness. However, excessive fastener density leads to localised stress concentrations. Among the configurations studied, the two rivet layout provides the most balanced response under quasi static loading.

High velocity impact behaviour (Case 2) is investigated using LS DYNA explicit dynamics on a GFRP Nomex sandwich panel, with facesheet core debonding modelled using a tiebreak contact formulation. A rigid spherical impactor with an initial velocity of 900 m/s is applied. The four rivet configuration significantly limits facesheet core debonding and reduces Nomex core crushing.

Overall, the results demonstrate that the optimal rivet configuration strongly depends on the loading regime. An intermediate rivet density is most effective for controlling quasi static delamination, whereas higher rivet density is required to mitigate impact induced damage in sandwich structures.

Chapter 1

Introduction

The aerospace industry widely adopts composite laminates and sandwich structures because of their high strength-to-weight ratio and structural efficiency. However, these materials are prone to interlaminar damage, such as delamination in laminates and facesheet–core debonding in sandwich panels, which can significantly reduce structural performance. Mechanical fastening is commonly employed to enhance load transfer and damage tolerance, however the introduction of rivets also generates local stress concentrations that strongly influence damage initiation and growth.

Numerical modelling provides an effective means of investigating these complex damage mechanisms. Cohesive zone modelling (CZM) has been extensively used to simulate progressive delamination in composite laminates under quasi-static loading, while explicit dynamic solvers, such as LS-DYNA, are well suited for analysing high-velocity impact events involving sandwich structures.

This study examines the effect of rivet configuration on interlaminar damage under different loading regimes through two numerical case studies. Case Study 1 uses ANSYS Composite PrepPost (ACP) to analyse quasi-static delamination behaviour in a 12-ply quasi-isotropic GFRP laminate with varying rivet layouts. Case Study 2 employs LS-DYNA explicit dynamics to investigate impact-induced face sheet–core debonding and core crushing in a GFRP–Nomex sandwich panel. The results provide insight into how rivet distribution influences damage tolerance under both static and dynamic loading conditions.

Chapter 2

Related Works

Extensive research has been conducted on damage mechanisms and joining strategies in composite laminates and sandwich structures, particularly focusing on delamination, facesheet–core debonding, and load transfer efficiency. Adhesive bonding remains the most common joining method in sandwich panels due to its ability to distribute loads uniformly; however, it is highly sensitive to manufacturing defects, impact damage, and environmental degradation.

Cohesive zone modeling (CZM) has been widely adopted to simulate interlaminar delamination in composite laminates and sandwich structures. Early work by Wang [1] demonstrated that cohesive elements can accurately capture facesheet–core interface delamination in honeycomb sandwich panels under static loading conditions. Subsequent studies by Wang et al. [2] further validated bilinear traction–separation laws for predicting peeling behavior in sandwich structures, reporting good agreement with experimental results. More recent developments by Ai et al. [3] and Himanshu and Ramaswamy [4] have extended CZM formulations to improve numerical stability and enable more accurate representation of non-self-similar delamination growth in composite systems.

Hybrid joining techniques that combine adhesive bonding with discrete mechanical reinforcement have gained increasing attention due to their improved damage tolerance. Studies by Yu et al. [5] and Kumar et al. [6] showed that friction stir spot welded–adhesive hybrid joints exhibit significantly enhanced peel and tensile–shear strength compared to purely adhesive joints. These improvements were attributed to redundant load paths provided by the mechanical reinforcement, which continue to carry load after adhesive failure. Review studies by Maggiore et al. [7] further highlighted the benefits of hybrid joining approaches for improving joint reliability in aerospace and automotive structures.

Research on sandwich panel impact behavior has identified facesheet–core debonding and core crushing as dominant failure modes under transverse loading. Jiang et al. [8] demonstrated that adhesive material properties and interface strength strongly influence damage development under edgewise compression, while Mohammed et al. [9] reviewed failure mechanisms in metal and composite sandwich panels, emphasizing the role of core density and interface integrity. These findings underline the importance of load-path redundancy in mitigating impact-induced damage.

Although significant progress has been made in modeling delamination and hybrid joints, most existing studies focus on coupon-level specimens, metallic systems, or a single loading regime. Limited work has investigated the combined influence of mechanical fastener configuration on both quasi-static delamination and high-velocity impact behavior in GFRP–Nomex honeycomb sandwich panels at the structural scale. The present study addresses this gap by numerically evaluating the effect of rivet number and layout under both static and dynamic loading conditions using complementary finite element approaches.

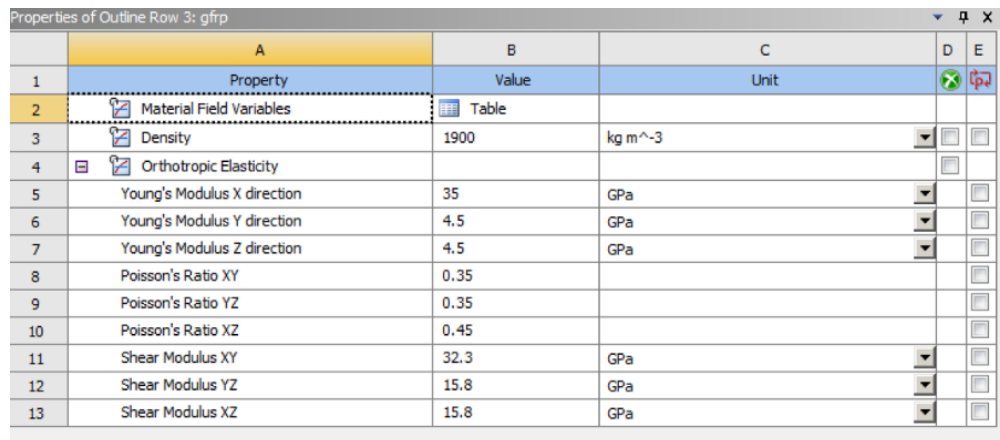
Chapter 3

Material Chacterization

The material properties used in this study are defined using the ANSYS Engineering Data library, where the properties for GFRP facesheets, Nomex honeycomb core, and epoxy adhesive are adopted from values reported in the literature for aerospace sandwich structures. These properties are selected to realistically represent the elastic and interfacial behaviour of composite sandwich panels under static and dynamic loading.

3.1 GFRP Properties

The GFRP facesheets are modelled as orthotropic elastic materials. Engineering properties defined in ANSYS include density, directional Young's moduli, Poisson's ratios, and shear moduli in the principal material directions. The longitudinal Young's modulus is assigned higher than the transverse moduli to capture fibre-dominated stiffness, while appropriate shear moduli are specified to represent in-plane and out-of-plane shear behaviour. These properties enable accurate simulation of load transfer, stress distribution, and interlaminar response within the composite laminate.



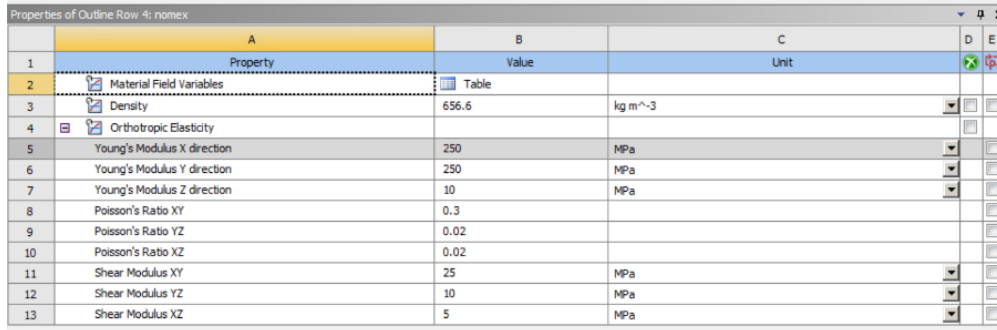
The screenshot shows a table titled 'Properties of Outline Row 3: gfrp'. The table has five columns: A (Property), B (Value), C (Unit), D, and E. Row 2 is highlighted in yellow and contains 'Material Field Variables' and 'Table'. Row 3 contains 'Density' with a value of 1900 and unit kg m^-3. Row 4 is highlighted in purple and contains 'Orthotropic Elasticity'. Rows 5-13 list various material properties with their values and units.

	A	B	C	D	E
1	Property	Value	Unit		
2	Material Field Variables	Table			
3	Density	1900	kg m^-3		
4	Orthotropic Elasticity				
5	Young's Modulus X direction	35	GPa		
6	Young's Modulus Y direction	4.5	GPa		
7	Young's Modulus Z direction	4.5	GPa		
8	Poisson's Ratio XY	0.35			
9	Poisson's Ratio YZ	0.35			
10	Poisson's Ratio XZ	0.45			
11	Shear Modulus XY	32.3	GPa		
12	Shear Modulus YZ	15.8	GPa		
13	Shear Modulus XZ	15.8	GPa		

Figure 3.1: GFRP Face sheet Lamina properties

3.2 Nomex Honeycomb Core Properties

The Nomex honeycomb core is represented as an orthotropic elastic material to account for its highly directional mechanical behaviour. The material definition includes density, Young's moduli in the thickness and in-plane directions, Poisson's ratios, and shear moduli. A lower out-of-plane modulus is specified to allow compressive deformation and core crushing under transverse loading, while relatively higher in-plane shear stiffness is maintained to ensure structural stability. This representation captures the key mechanical response of the honeycomb core without explicitly modelling individual cells.

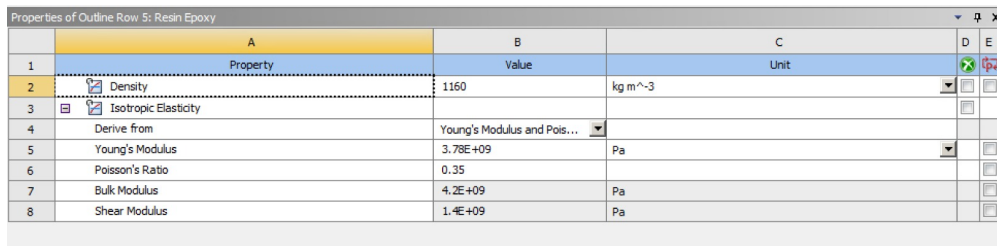


Properties of Outline Row 4: nomex				
	A	B	C	
	Property	Value	Unit	
2	Material Field Variables	Table		
3	Density	656.6	kg m ⁻³	
4	Orthotropic Elasticity			
5	Young's Modulus X direction	250	MPa	
6	Young's Modulus Y direction	250	MPa	
7	Young's Modulus Z direction	10	MPa	
8	Poisson's Ratio XY	0.3		
9	Poisson's Ratio YZ	0.02		
10	Poisson's Ratio XZ	0.02		
11	Shear Modulus XY	25	MPa	
12	Shear Modulus YZ	10	MPa	
13	Shear Modulus XZ	5	MPa	

Figure 3.2: Nomex Honeycomb Core Material Properties

3.3 Epoxy Adhesive Properties

The epoxy adhesive layer is modelled using isotropic elastic properties defined within the ANSYS Engineering Data module. The adhesive material properties include density, Young's modulus, and Poisson's ratio, which govern stiffness and deformation prior to damage initiation. These properties provide the elastic response of the bonding layer, while failure and separation behaviour are governed separately through the cohesive zone model and interface definitions. This approach allows realistic simulation of adhesive load transfer and interfacial degradation.



Properties of Outline Row 5: Resin Epoxy				
	A	B	C	
	Property	Value	Unit	
2	Density	1160	kg m ⁻³	
3	Isotropic Elasticity			
4	Derive from	Young's Modulus and Pois...		
5	Young's Modulus	3.78E+09	Pa	
6	Poisson's Ratio	0.35		
7	Bulk Modulus	4.2E+09	Pa	
8	Shear Modulus	1.4E+09	Pa	

Figure 3.3: Epoxy Adhesive Material Properties

3.4 Cohesive Zone Model (CZM) Properties

The cohesive interface properties, including normal and shear strengths and Mode I and Mode II fracture toughness values, were defined using the ANSYS Engineering Data module

based on literature-reported values for quasi-isotropic GFRP laminates. Interface degradation was controlled by a scalar damage variable ranging from zero (undamaged) to one (complete decohesion), enabling simulation of progressive delamination growth under static loading.

Properties of Outline Row 3: cm				
	A	B	C	
1	Property	Value	Unit	
2	Material Field Variables	Table		
3	Isotropic Elasticity			
4	Derive from	Young's Modulus and Poisson's ...		
5	Young's Modulus	30000	MPa	
6	Poisson's Ratio	0.3		
7	Bulk Modulus	2.9E+10	Pa	
8	Shear Modulus	1.1538E+10	Pa	
9	Bilinear for Interface Delamination			
10	Maximum Normal Traction	35	MPa	
11	Normal Displacement Jump at Completion of Debonding	0.017	mm	
12	Maximum Tangential Traction	60	Pa	
13	Tangential Displacement Jump at Completion of Debonding	0.03	mm	
14	Ratio	0.5		
15	Non-Dimensional Weighting Parameter	1.7		

Figure 3.4: Cohesive Zone Model Parameters

Chapter 4

Methodology

4.1 Setup

This study conducted through two independent finite element case studies to investigate damage initiation, propagation and load re-distributions in GFRP-Nomex honeycomb sandwich panels subjected to quasi-static and high-velocity transverse loading. Quasi-isotropic ply interlaminar delamination is modelled using ACP, where cohesive zone modelling is embedded within a static structural analysis setup to capture damage initiation and propagation. High velocity transverse impact response is analysed separately using LS-DYNA explicit Dynamics, in which tiebreak contact techniques and composite damage modelling are used to depict face sheet, core separation and compact induced debonding. Maintain consistent comparison of damage trends between static and dynamic situations, same baseline geometries, rivet configurations, and material definitions are used even if the analyses are carried out using various solvers and formulations appropriate for their respective loading conditions.

4.2 Case Study I - ANSYS ACP Static Structural Analysis

4.2.1 Composite Laminate Architecture and Stack-Up Definition

A quasi-isotropic composite laminate was modelled using ANSYS Composite PrepPost (ACP). Laminate consisted of a 12-ply symmetric stacking sequence, designed to exhibit quasi-isotropic in-plane behaviour under static loading.

To enable the investigation of interlaminar delamination while preserving structural symmetry, a single cohesive interface was introduced at the laminate mid-plane between plies 6 and 7. This interface location ensures geometric and material symmetry and allows controlled initiation and propagation of delamination under applied loads.

The ACP-to-solver workflow followed a sequential coupling strategy, wherein the laminate definition and ply-level properties were established in ACP and subsequently transferred to

[illegible]

4.2.2 Cohesive Zone Model

Damage Initiation Criterion

$$\max \left\{ \left(\frac{\langle \sigma_n \rangle}{\sigma_{n,\max}} \right)^2, \left(\frac{\langle \tau_t \rangle}{\tau_{t,\max}} \right)^2, \left(\frac{\langle \tau_s \rangle}{\tau_{s,\max}} \right)^2 \right\} = 1 \quad (4.1)$$

- σ_n is the normal traction,
- τ_t and τ_s are the tangential shear tractions,
- $\sigma_{n,\max}$, $\tau_{t,\max}$, and $\tau_{s,\max}$ are the corresponding maximum cohesive strengths,
- The Macaulay brackets $\langle \cdot \rangle$ ensure that normal tractions contribute only under tensile loading, while shear tractions contribute through their absolute values.

Following damage initiation, delamination propagation was governed by the Benzeggagh–Kenane (BK) mixed-mode fracture criterion, defined as:

$$G_{eq} = G_{IC} + (G_{IIC} - G_{IC}) \left(\frac{G_{II}}{G_I + G_{II}} \right)^\alpha \quad (4.2)$$

where:

- G_I and G_{II} are the Mode I and Mode II energy release rates, respectively,
- G_{IC} and G_{IIC} are the critical fracture energies for Modes I and II,
- α is the mixed-mode interaction parameter.

A value of $\alpha = 1.5$ was adopted, consistent with literature-reported values for GFRP inter-laminar behaviour.

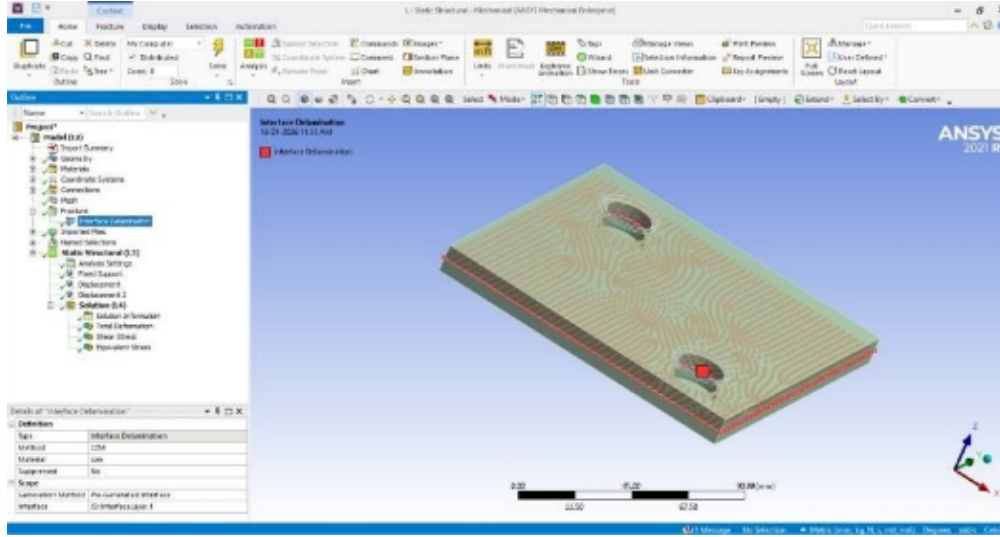


Figure 4.2: CZM Interface Delamination

The degradation of interface stiffness was controlled by a scalar damage variable $d \in [0, 1]$, where:

- $d = 0$ corresponds to the undamaged elastic response,
- $d = 1$ represents complete decohesion of the interface.

Cohesive material parameters were selected based on validated literature values for quasi-isotropic composite laminates.

4.2.3 Boundary Conditions and Loading Criteria

Two Loading Conditions were applied to investigate both unstable and stable delamination behaviour:

1. Force-controlled loading

A uniform tensile load of 20 kN was applied to the opposite faces of the laminate to induce peak interlaminar stresses and promote unstable delamination initiation.

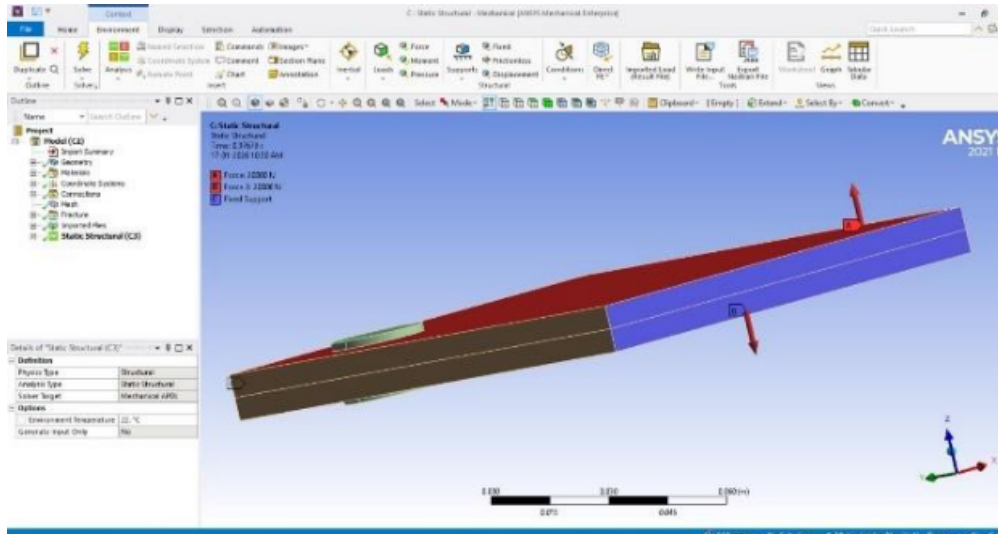


Figure 4.3: Force-Controlled Loading Boundary Conditions

2. Displacement-controlled loading

A remote displacement of 5 mm was applied to enable stable crack propagation and facilitate the extraction of delamination resistance (R-curve) behaviour.

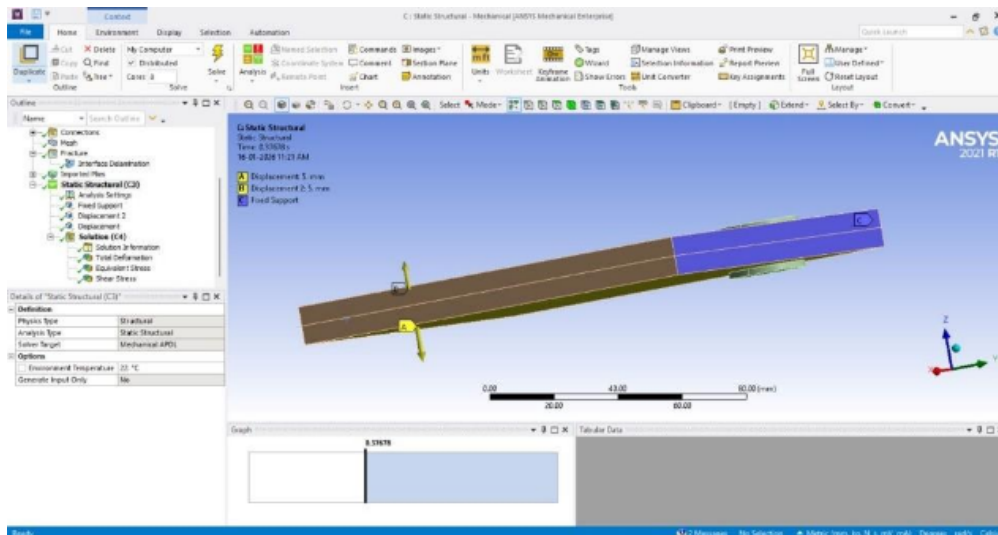


Figure 4.4: Displacement-Controlled Loading Setup

Boundary Conditions

To simulate fixed-support conditions, the composite panel was fully constrained at both lateral edges, restricting all translational degrees of freedom.

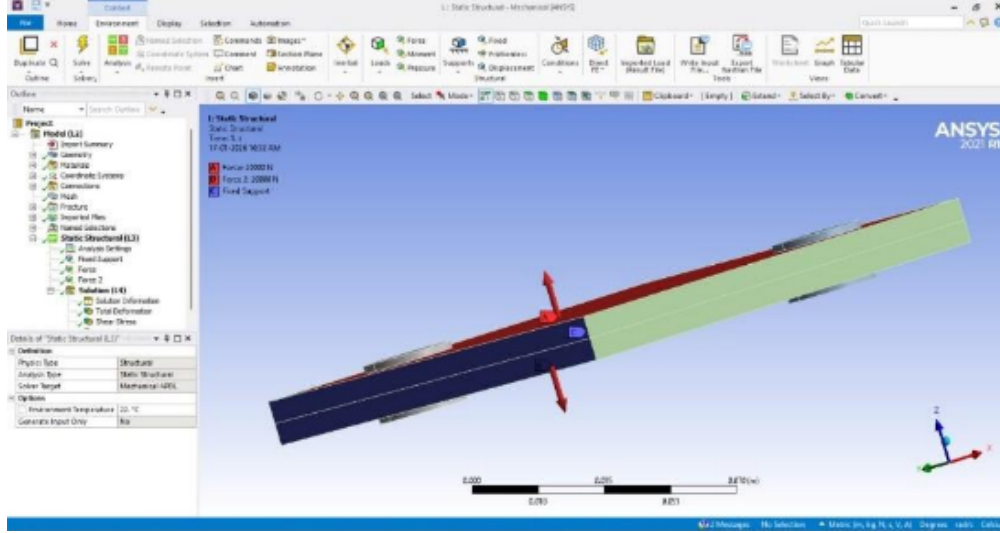


Figure 4.5: Fixed Support Boundary Conditions

4.2.4 Mesh Discretization

The composite laminate was discretized using three-dimensional hex-dominant solid elements. Local mesh refinement was applied near the cohesive mid-plane interface and around the rivet locations to accurately capture interlaminar stress gradients and delamination initiation. The in-plane element size in the vicinity of the cohesive interface was selected to be smaller than the estimated cohesive zone length, ensuring that multiple elements spanned the process zone and thereby improving the accuracy of delamination prediction. The cohesive interface was aligned with adjacent solid elements to ensure compatibility with the zero-thickness cohesive formulation.

4.2.5 Delamination Assessment Metrics

Delamination initiation and propagation were evaluated using output variables associated with the cohesive interface elements. The primary indicator of interlaminar damage was the **scalar cohesive damage variable**, which ranges from zero (undamaged interface) to unity (complete decohesion).

In addition, interlaminar shear stress distributions were extracted along the cohesive interface to assess the influence of rivet configuration on delamination-driving stresses. The spatial extent and distribution of damaged cohesive regions were used to qualitatively compare delamination growth among the different rivet layouts.

Global structural response parameters, including maximum von Mises stress and total deformation, were also monitored to correlate delamination behaviour with overall load redistribution and stiffness characteristics of the laminate.

4.3 Case Study II - LS-DYNA Impact Analysis

This case study investigates the influence of **rivet configuration** on **impact-induced facesheet–core debonding and core crushing** in composite sandwich panels using **LS-DYNA**. The objective is to quantify how **1-, 2-, and 4-rivet layouts** modify the damage footprint at the **facesheet–core interface** and within the **Nomex honeycomb core** under high-velocity transverse impact.

High-speed spherical impact generates **severe transient stress waves**, leading to localized bending of the facesheets, interfacial separation, and progressive core crushing. These highly dynamic phenomena were captured using an explicit dynamic solution strategy, which is well suited for short-duration, high-strain-rate events.

4.3.1 Numerical Model and Mesh Strategy

The sandwich panel was modelled using **three-dimensional solid elements** for both the composite facesheets and the honeycomb core. To maintain computational efficiency while accurately resolving damage near the impact region, a graded mesh strategy was adopted.

A locally refined mesh was employed in the vicinity of the impact zone and along the facesheet–core interface, where high stress gradients and damage initiation are expected. Away from the impact region, the mesh was progressively coarsened, with element sizes increasing to approximately 4 mm at the far-field boundaries. This mesh transitioning approach reduces overall model size while preserving numerical accuracy near critical regions.

4.3.2 Facesheet–Core Interface Modelling

Bonding between the facesheets and the Nomex core was represented using **surface-based tiebreak contact** in LS-DYNA. This formulation allows progressive debonding based on combined normal and shear failure criteria.

The normal and shear failure thresholds for the tiebreak interface were set to:

- Normal stress limit: 100 Pa
- Shear stress limit: 50 Pa

Initially, the interface nodes are fully tied, representing an intact bond. Once the stress thresholds are exceeded, the interface progressively degrades, enabling simulation of facesheet–core debonding under impact loading.

4.3.3 Boundary Conditions and Loading Criteria

Impactor Definition

A rigid steel spherical impactor was used to simulate a severe transverse impact event representative of runway debris or high-speed foreign object damage (FOD). The impactor was

modelled as a rigid body to ensure that its deformation remained negligible relative to that of the sandwich panel.

- Sphere radius: 5 mm
- Initial velocity: 900 m/s
- Impact direction: Normal to the panel surface
- Impact location: Centre of the panel

To simulate fixed-support conditions, the sandwich panel was fully constrained along both lateral edges, restricting all translational degrees of freedom. This boundary condition ensures that the impact response is dominated by local deformation, interfacial failure, and core crushing.

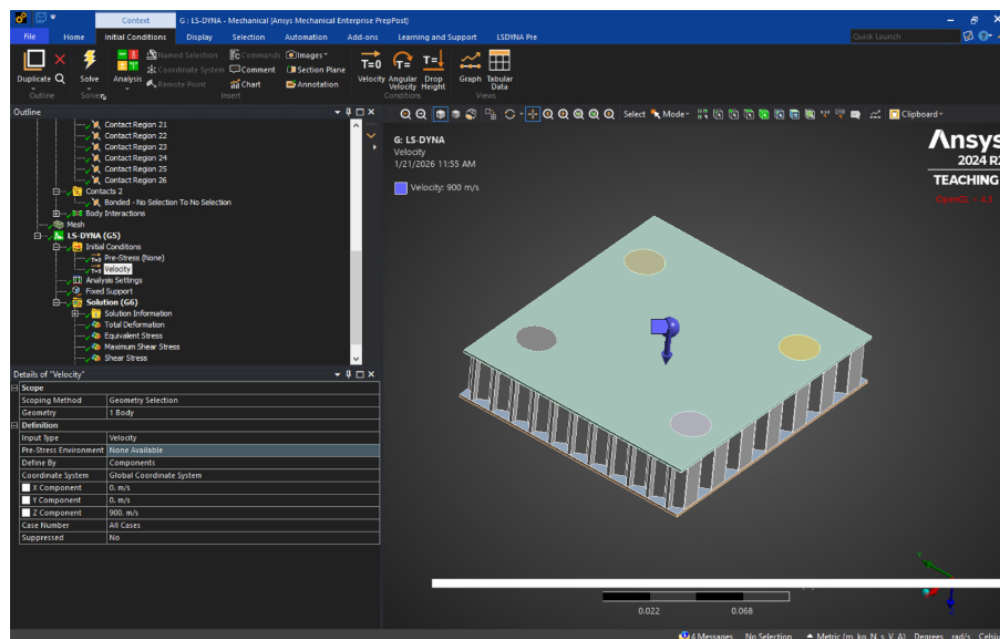


Figure 4.6: Rigid Spherical Impactor and Initial Velocity Definition

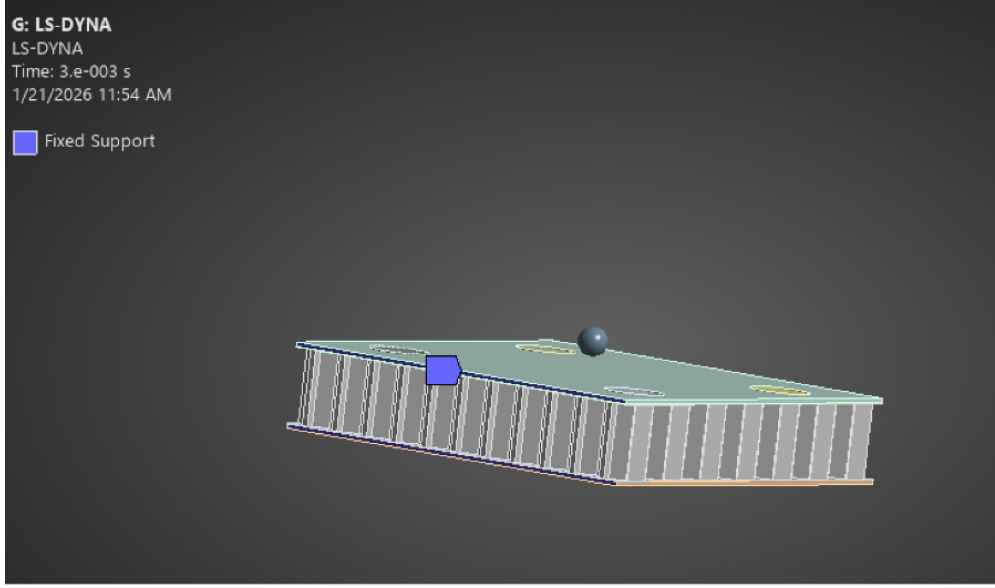


Figure 4.7: Fixed Support Boundary Conditions - LS-DYNA Impact Model

4.3.4 Delamination Assessment Metrics

Delamination at the facesheet-core interface was assessed using the tiebreak contact formulation in LS-DYNA. Interface degradation was quantified through the nodal contact gap damage variable, which ranges from 0 for a fully bonded interface to 1 for complete debonding. This variable served as the primary indicator of delamination initiation, growth, and final extent. Delamination initiation was defined as the first occurrence of a damage value exceeding a small threshold ($d \geq 0.05$), while complete delamination was identified when the damage variable reached unity ($d = 1$).

Chapter 5

Results and Analysis

5.1 Case Study 1: Static Delamination Response of Riveted Composite Laminates

A static structural analysis was performed using ANSYS Composite PrepPost (ACP) coupled with a bilinear cohesive zone modelling (CZM) formulation to investigate delamination behaviour in a 12-ply quasi-isotropic composite laminate. Three mechanical fastening configurations: 1-rivet, 2-rivet, and 4-rivet layouts were analysed under an identical fixed static load. The study focused on the influence of rivet count on global structural response and interlaminar damage development. No impact loading was considered in this case study.

5.1.1 Global Displacement–Load Response and Force–Load Response

The global structural response of the laminate was evaluated under both force controlled (load–displacement) and displacement controlled (force–displacement) loading conditions using maximum von Mises stress and total deformation as primary indicators. These metrics provide insight into global stiffness, load transfer efficiency and stress redistribution associated with different rivet configurations.

Under force controlled loading, the single rivet configuration exhibited the highest total deformation, indicating reduced global stiffness and significant load concentration at the fastener laminate interface. The elevated deformation response suggests inefficient load sharing, resulting in higher stress localization around the rivet. Introducing a second rivet led to a clear improvement in structural behaviour. The two rivet configuration showed a notable reduction in both maximum von Mises stress and total deformation compared to the single rivet case, demonstrating improved load redistribution and a more uniform stress field within the laminate.

The four rivet configuration resulted in the lowest total deformation under force controlled loading, confirming the expected increase in global stiffness with increasing fastener count.

However, this stiffness enhancement was accompanied by an increase in local stress levels, indicating stronger interaction effects between neighbouring fasteners and the onset of stress concentration due to reduced spacing.

A similar trend was observed under displacement controlled loading. For a prescribed displacement, the single rivet configuration developed the highest equivalent and shear stresses, reflecting higher force demand on the laminate. The two rivet configuration reduced peak stress levels significantly, indicating more effective force sharing between fasteners. While the four rivet configuration further increased load carrying capacity, it also produced higher local stresses, suggesting diminishing returns in damage mitigation due to fastener interaction effects.

Overall, the results indicate that the two rivet configuration provides the most balanced global response under both loading regimes, achieving effective stiffness enhancement and stress redistribution without introducing excessive local stress concentrations.

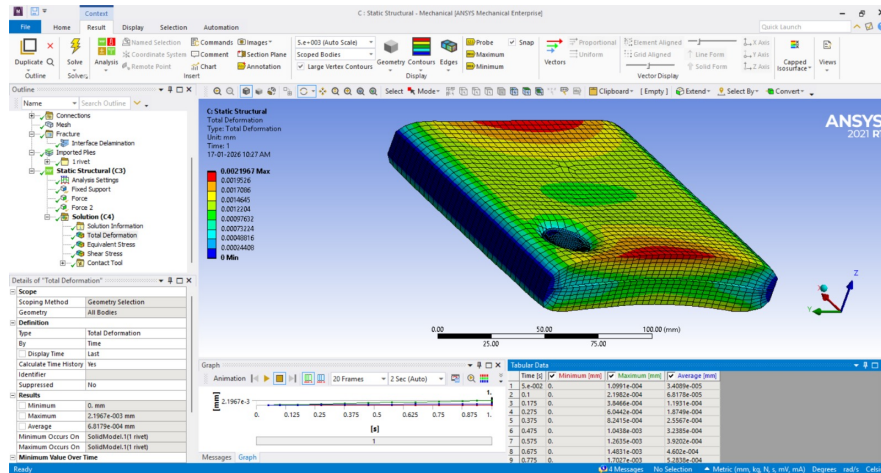


Figure 5.1: Total deformation for 1-rivet condition for Force–Load Response

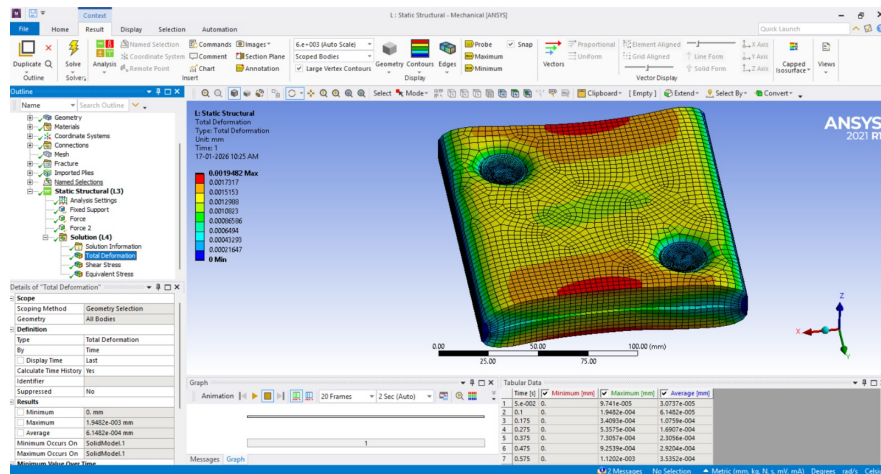


Figure 5.2: Total deformation for 2-rivet condition for Force–Load Response

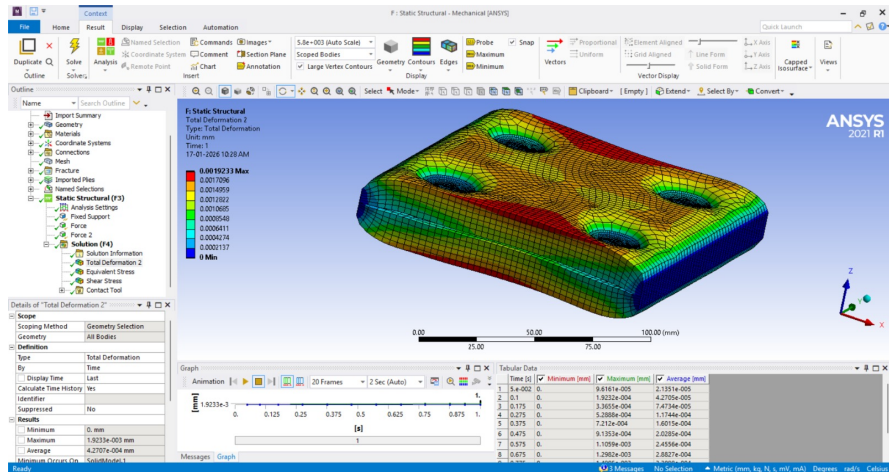


Figure 5.3: Total deformation for 4-rivet condition for Force-Load Response

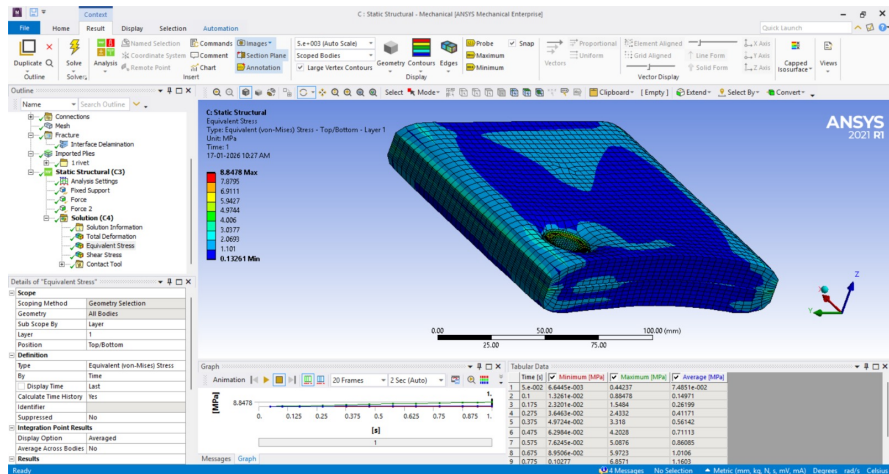


Figure 5.4: Equivalent stress for 1-rivet condition for Force-Load Response

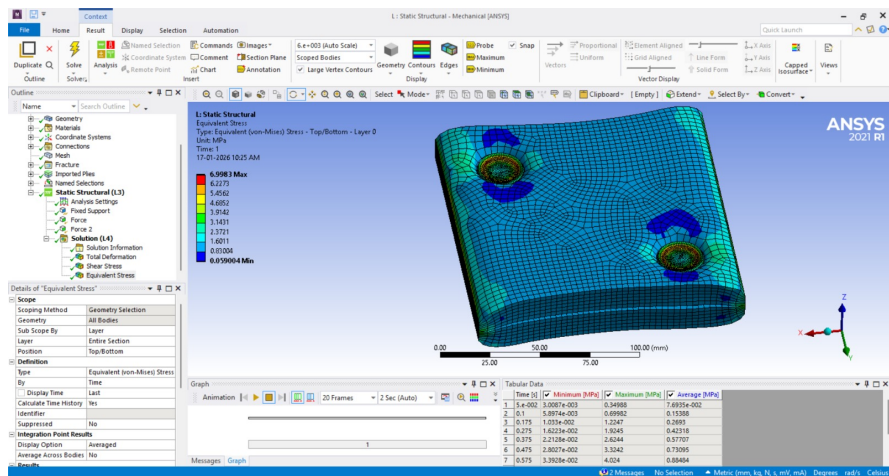


Figure 5.5: Equivalent stress for 2-rivet condition for Force-Load Response

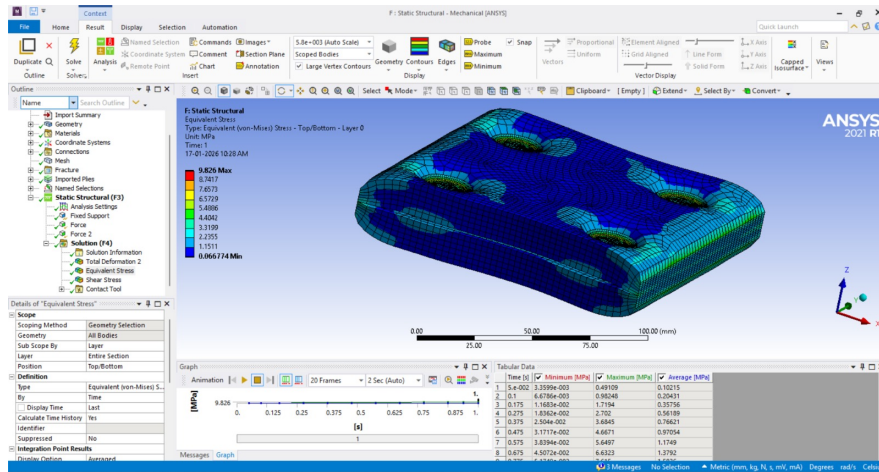


Figure 5.6: Equivalent stress for 4-rivet condition for Force–Load Response

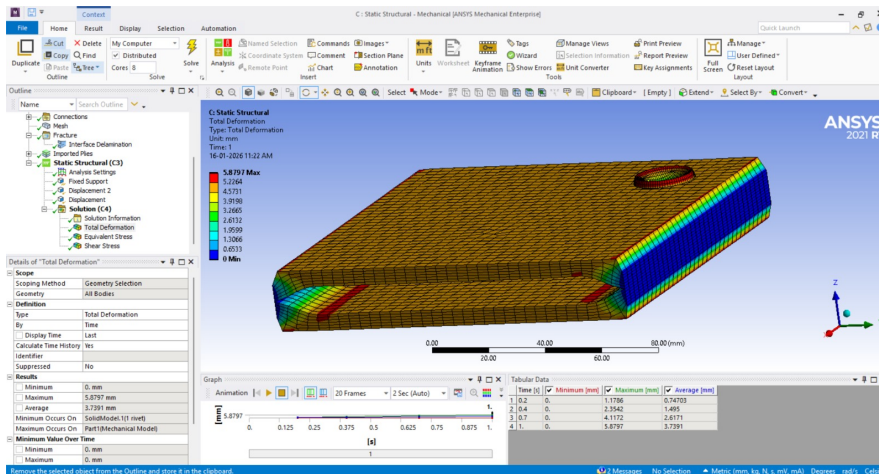


Figure 5.7: Total deformation for 1-rivet condition for Displacement–Load Response

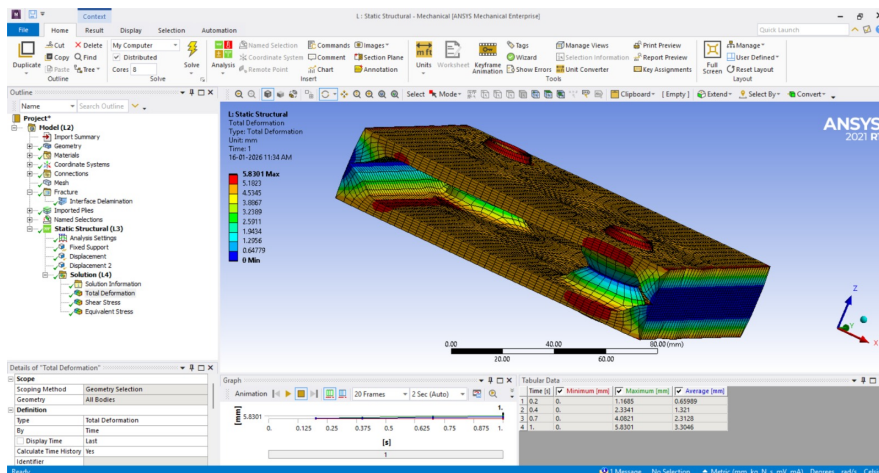


Figure 5.8: Total deformation for 2-rivet condition for Displacement–Load Response

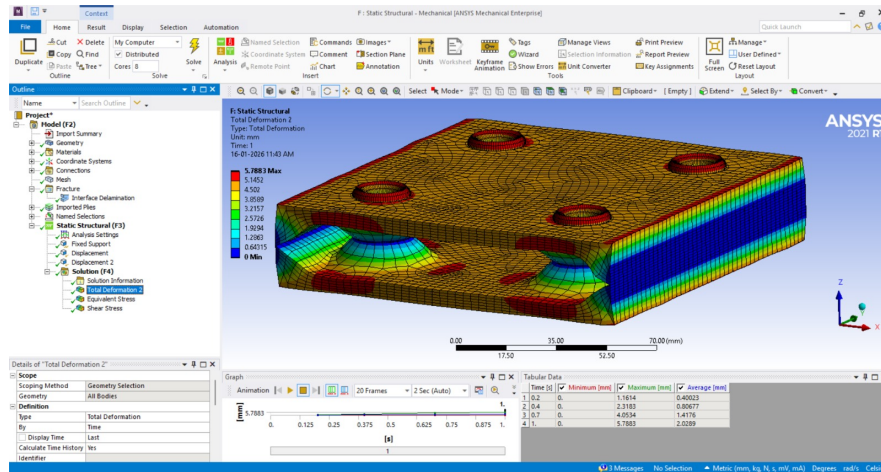


Figure 5.9: Total deformation for 4-rivet condition for Displacement–Load Response

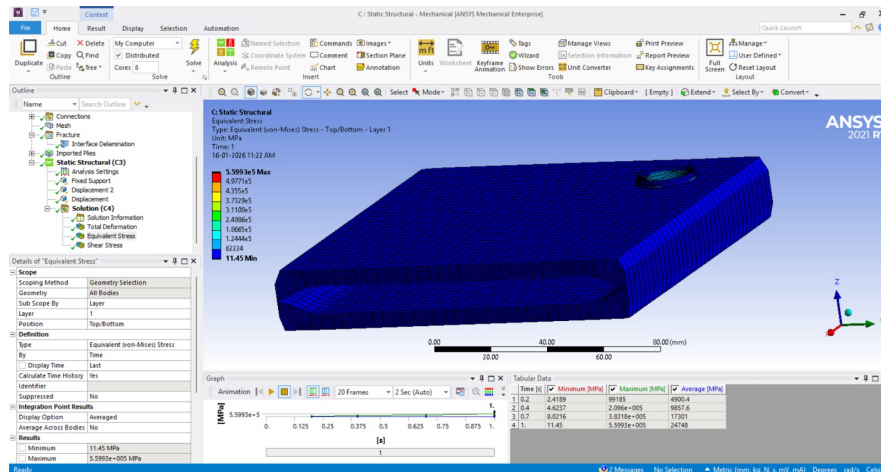


Figure 5.10: Equivalent stress for 1-rivet condition for Displacement–Load Response

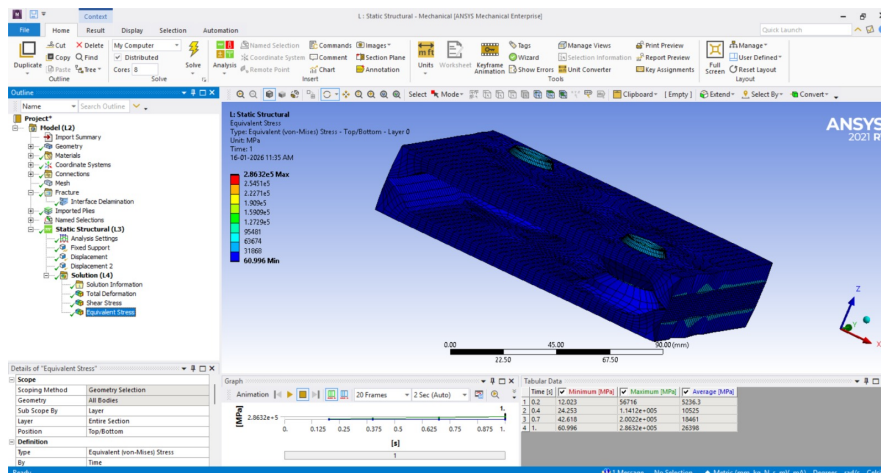


Figure 5.11: Equivalent stress for 2-rivet condition for Displacement–Load Response

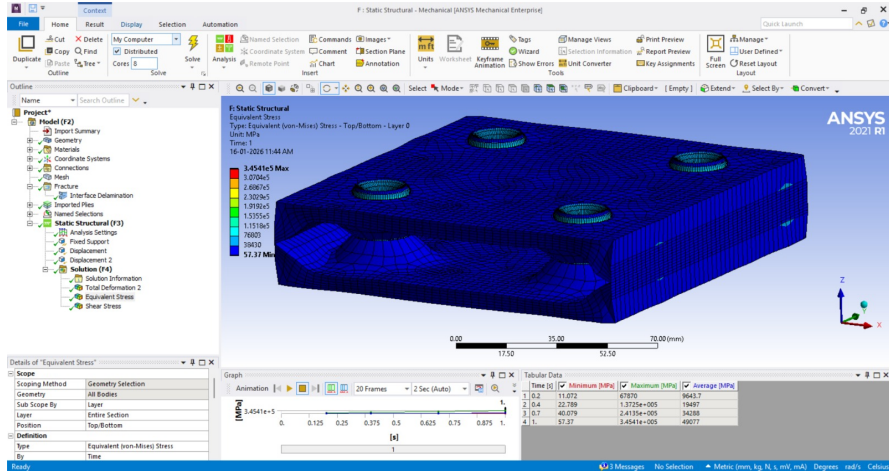


Figure 5.12: Equivalent stress for 4-rivet condition for Displacement–Load Response

Rivet Configuration	Max von Mises Stress (MPa)	Total Deformation (mm)	Max Shear Stress (MPa)	Min Shear Stress (MPa)	Shear Stress Range (MPa)
1 Rivet	8.85	0.002197	+1.401	−1.550	2.951
2 Rivets	7.00 (−21%)	0.001948 (−11%)	+1.429 (+2%)	−1.470 (+5%)	2.899
4 Rivets	9.83 (+11%)	0.001923 (−12%)	+1.595 (+14%)	−1.599 (+3%)	3.194

Table 5.1: Comparison of Stress and Deformation Results

Tables 5.1 and 5.2 summarize the global structural response in terms of stress and deformation for constant force and constant displacement loading conditions across different rivet configurations.

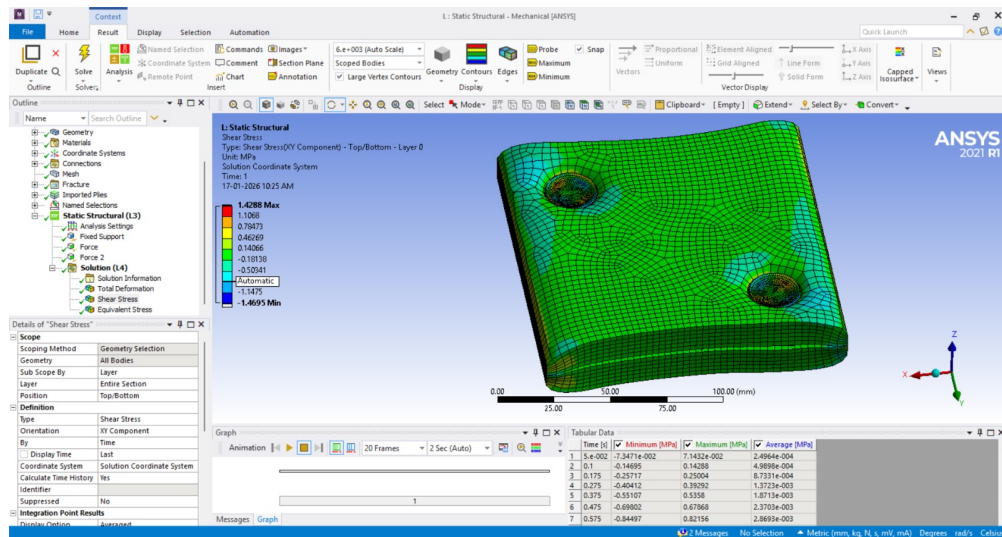
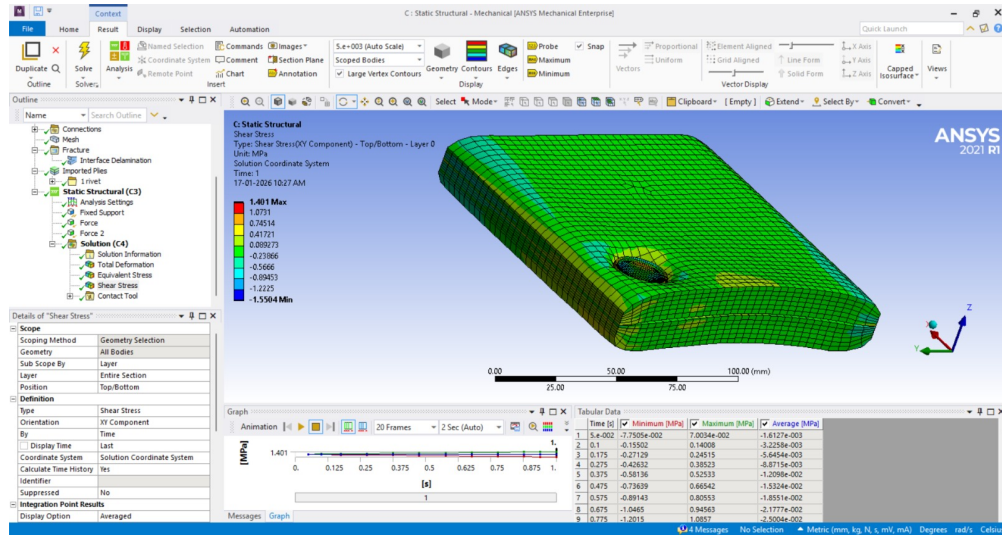
Rivet Configuration	Max Deformation (mm)	Max Equivalent Stress (MPa)	Max Shear Stress (MPa)	Min Shear Stress (MPa)
1 Rivet	5.8797	5.5993×10^5	39,852	−39,706
2 Rivets	5.8301	2.8632×10^5	44,756	−44,822
4 Rivets	5.7883	3.4541×10^5	51,193	−51,179

Table 5.2: Summary of Deformation and Stress Analysis

5.1.2 Interlaminar Shear Stress Distribution

Interlaminar shear stresses extracted from the cohesive interfaces provide insight into delamination driving mechanisms. The 2-rivet configuration demonstrated a relatively uniform shear stress distribution with a reduced stress range compared to the 1-rivet case, indicating more balanced interlaminar load transfer.

In contrast, the 4-rivet configuration exhibited the highest interlaminar shear stress range. This increase is attributed to fastener interaction effects, which generate steeper stress gradients within the laminate and elevate local interfacial loading.



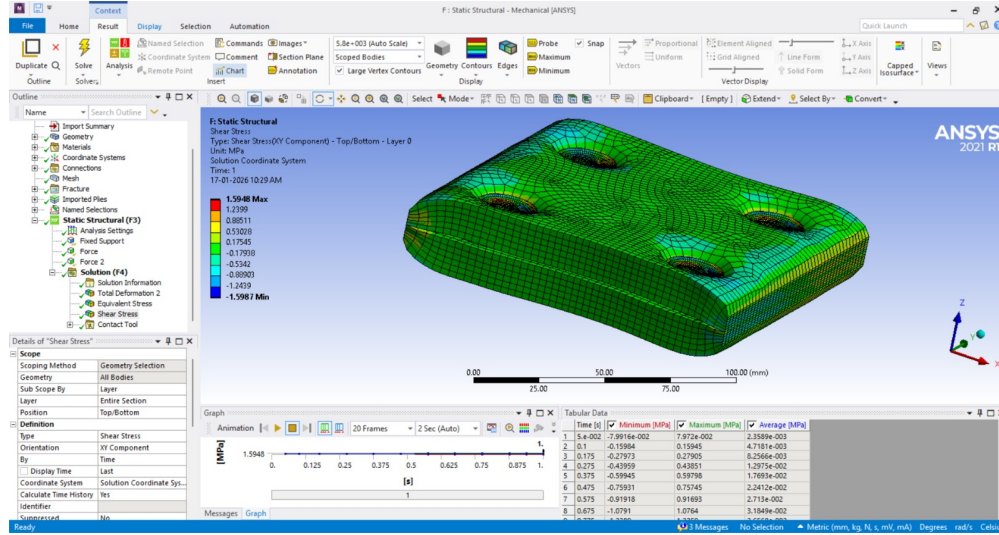


Figure 5.15: Shear Stress 3

5.1.3 Cohesive Damage and Delamination Behaviour

Delamination evolution was evaluated using cohesive damage variables associated with the CZM interfaces. The 1-rivet configuration showed localized regions of higher damage intensity concentrated near the fastener, consistent with higher deformation and stress concentration.

The 2-rivet configuration exhibited a more distributed and less intense damage pattern, indicating improved resistance to delamination growth under static loading. The cohesive response suggests that load redistribution between fasteners effectively reduces localized interlaminar damage.

The 4-rivet configuration, despite its reduced global deformation, showed localized increases in cohesive damage near fastener clusters. This behaviour reflects the combined effects of increased stiffness and elevated interlaminar shear demands in regions of fastener interaction.

5.1.4 Comparative Performance of Rivet Configurations

The results demonstrate that increasing rivet count enhances global stiffness and reduces overall deformation; however, excessive fastener density can introduce localized stress amplification and increased interlaminar loading. Among the configurations examined, the 2-rivet arrangement provides the most balanced performance, combining reduced global stress, moderate deformation, and controlled delamination development.

5.2 Case Study 2: LS-DYNA Impact Analysis Results

This section presents the results of the explicit dynamic impact simulations conducted using LS-DYNA to evaluate the influence of rivet configuration on facesheet–core debonding and Nomex core crushing in a composite sandwich panel subjected to high-velocity transverse

impact. Three fastening layouts: 1-rivet, 2-rivet, and 4-rivet configurations were analysed under identical impact and boundary conditions.

5.2.1 Global Impact Response

All configurations exhibited a highly transient response characterized by rapid facesheet bending, stress wave propagation through the sandwich thickness, and localized core deformation beneath the impactor. The rigid spherical impactor generated a short-duration, high-magnitude contact force, followed by oscillatory force decay associated with elastic wave reflections and progressive damage evolution.

While the overall impact duration was similar across all configurations, differences in peak contact force and force decay characteristics were observed, indicating variations in load transfer efficiency and energy dissipation mechanisms associated with rivet distribution.

5.2.2 Facesheet–Core Debonding Behaviour

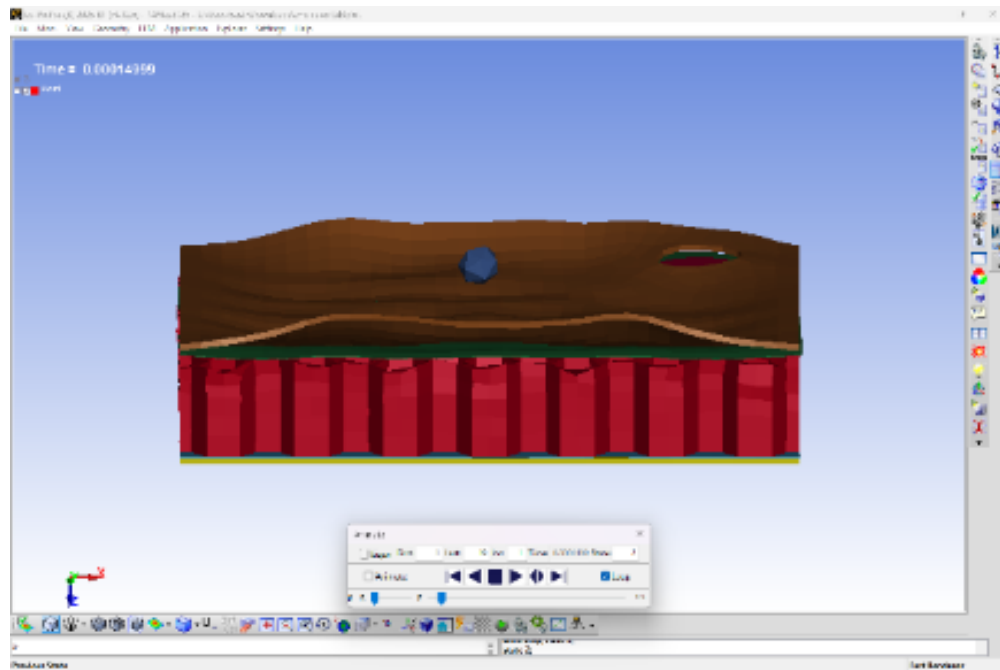


Figure 5.16: Debonding Evolution of single rivet

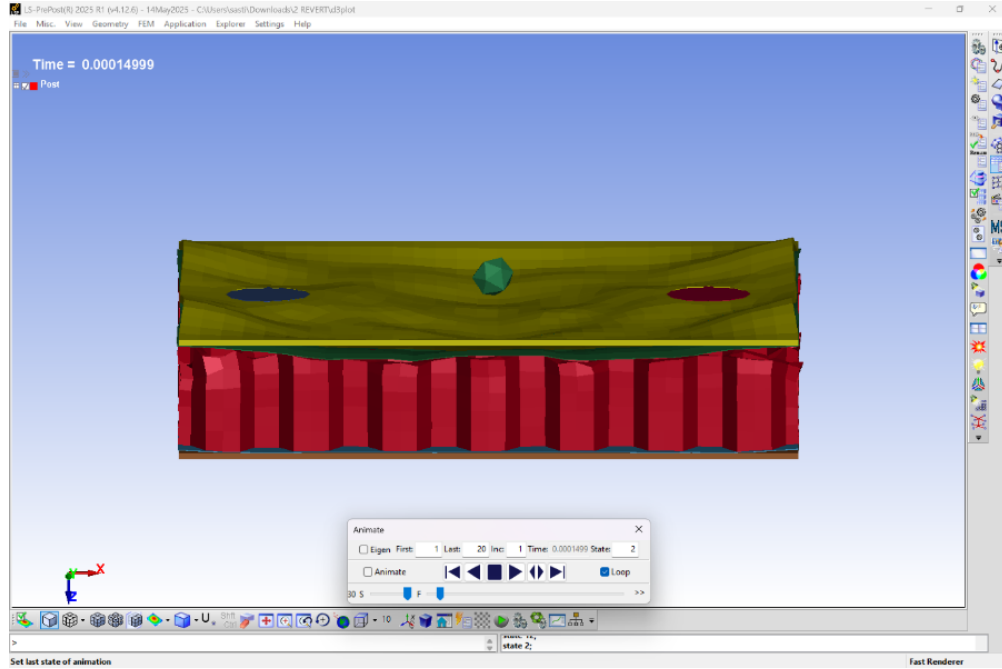


Figure 5.17: Debonding Evolution of 2 riverts

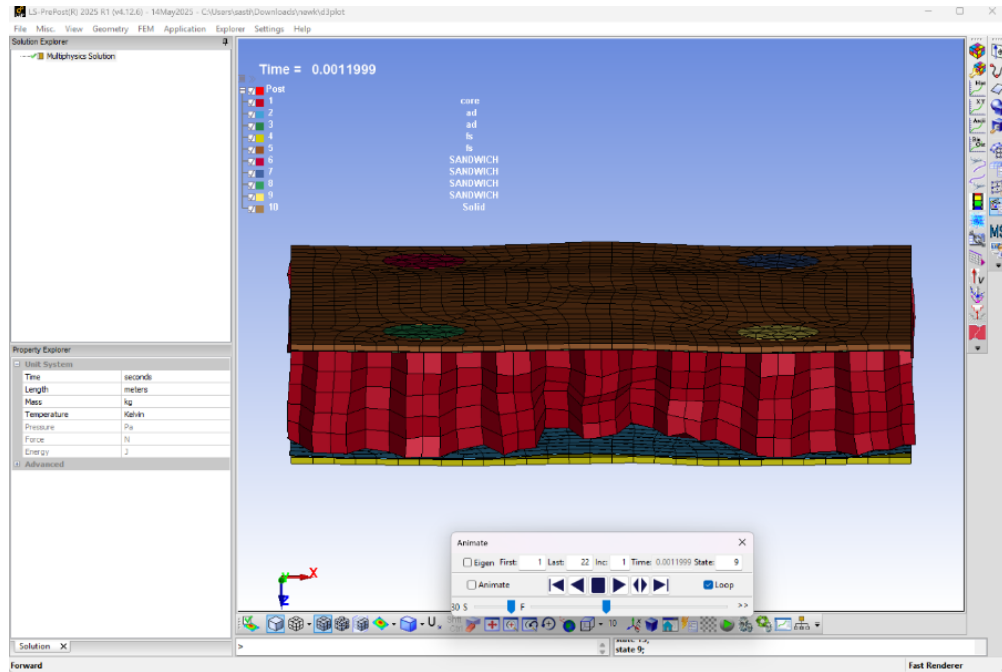


Figure 5.18: Debonding Evolution of 4 riverts

Facesheet–core debonding was quantified using the nodal contact gap parameter ξ , extracted from the tiebreak contact interface. Regions where $\xi = 1$ correspond to complete interface separation, while intermediate values indicate partial degradation.

1-rivet configuration: Extensive debonding developed rapidly around the impact zone and

propagated outward along the facesheet–core interface. Large contiguous regions reached $\xi = 1$, indicating widespread loss of interfacial integrity. Approximately 70% of the interface area experienced complete debonding, reflecting limited load redistribution capability in the single-rivet layout.

2-rivet configuration: The introduction of a second rivet reduced the extent of complete debonding and promoted more distributed damage patterns. Fully debonded regions were more localized near the impact zone, with surrounding partially damaged regions indicating gradual interface degradation. This configuration demonstrated improved load sharing compared to the single-rivet case.

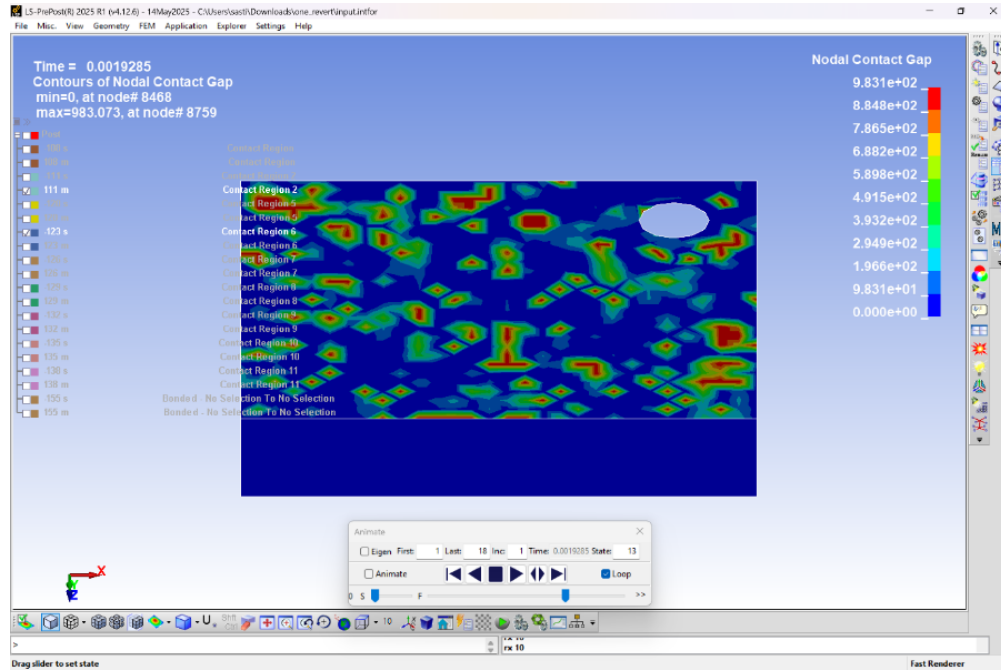


Figure 5.19: Nodal Contact Gap Contours - 1-Rivet Configuration

4-rivet configuration: The four-rivet layout exhibited the most confined debonding footprint. Fully debonded regions were restricted to a smaller area directly beneath the impactor, while large portions of the interface remained intact or only partially damaged. This indicates effective redistribution of impact-induced loads through multiple load paths.

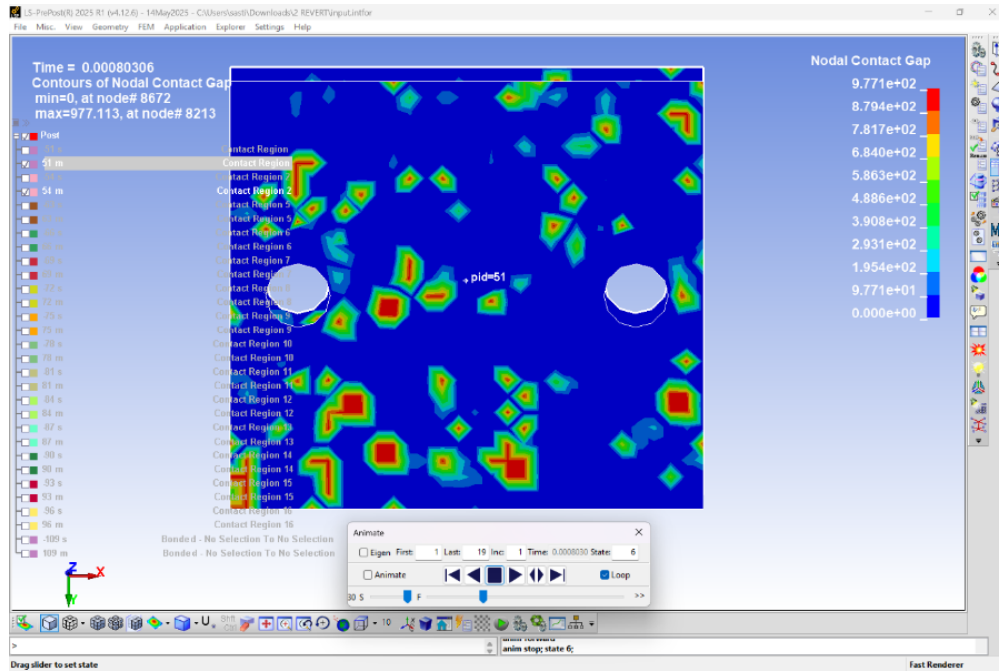


Figure 5.20: Nodal Contact Gap Contours - 2-Rivet Configuration

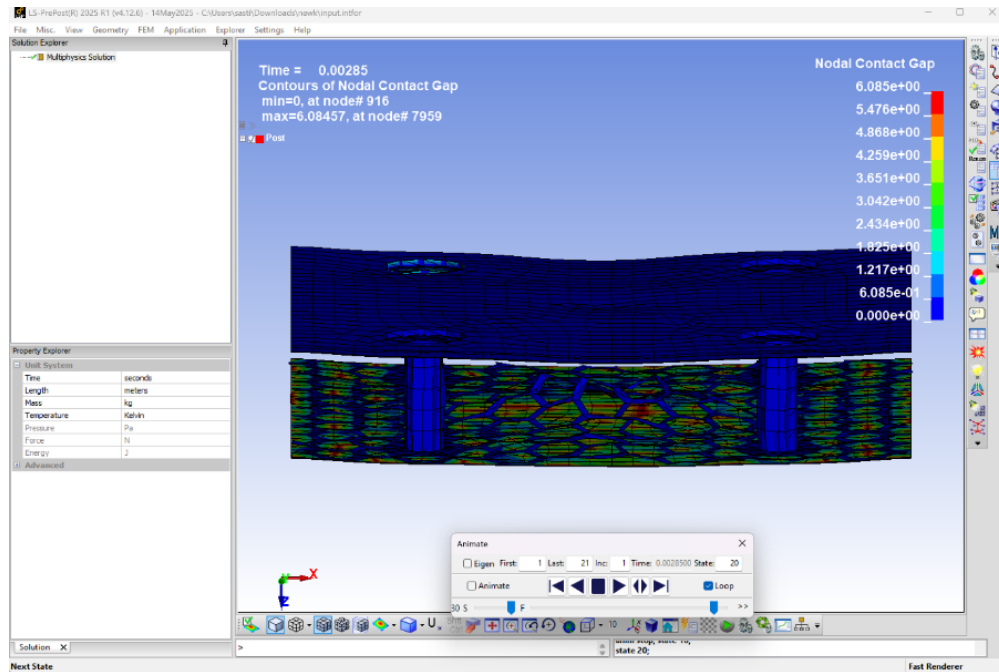


Figure 5.21: Nodal Contact Gap Contours - Facesheet Core Debonding

5.2.3 Nomex Core Crushing

Core crushing was evaluated using the out-of-plane compressive strain within the Nomex honeycomb core. Regions exceeding a prescribed compressive strain threshold were integrated to quantify the volume of significantly crushed core material.

- The 1-rivet configuration showed extensive core crushing beneath the impactor, consistent with high localized bending and limited restraint of the facesheets.
- The 2-rivet configuration reduced the volume of crushed core, indicating improved constraint and energy dissipation.
- The 4-rivet configuration provided the greatest resistance to core crushing, reducing the damaged core volume by more than 50% compared to the single-rivet case.

5.2.4 Impact Force-Time Response

The impact force–time histories reveal distinct differences in energy management between configurations. The 1-rivet configuration exhibited a sharp peak force followed by rapid force decay, consistent with localized failure and limited structural engagement. In contrast, the 4-rivet configuration produced a broader force–time response, indicating more effective load redistribution and enhanced energy absorption. The 2-rivet configuration showed intermediate behaviour.

5.2.5 Comparative Assessment of Rivet Configurations

A comparative evaluation of the three rivet layouts highlights clear trends:

- Increasing rivet count improves load-path redundancy, reducing both facesheet–core debonding and core crushing.
- The single-rivet configuration is most susceptible to widespread interface failure and severe core damage.
- The two-rivet configuration provides moderate improvement but remains sensitive to localized impact damage.
- The four-rivet configuration consistently demonstrates superior damage containment, reduced debonded area, and enhanced core protection.

These trends confirm that rivet distribution plays a critical role in governing inertia-driven damage mechanisms in sandwich structures.

Chapter 6

Conclusion

This study investigated the influence of rivet configuration on interlaminar damage behaviour in composite laminates and sandwich panels under quasi-static and high-velocity impact loading. Numerical simulations were carried out using ANSYS Composite PrepPost with cohesive zone modelling for static delamination analysis and LS-DYNA explicit dynamics for impact-induced facesheet–core debonding and core crushing.

Under quasi-static tensile loading, increasing rivet count improved global stiffness and load redistribution. While the four-rivet configuration produced the lowest overall deformation, it also generated higher localized interlaminar stresses due to fastener interaction. The two-rivet configuration provided the most balanced performance, achieving reduced stress levels, controlled delamination growth, and moderate deformation.

Under high-velocity impact loading, damage behaviour was dominated by inertia-driven mechanisms. In this regime, increased load-path redundancy was critical. The four-rivet configuration significantly reduced facesheet–core debonding and limited Nomex core crushing more effectively compared to the single-rivet case.

The results demonstrate that rivet configuration should be selected based on the dominant loading regime. Intermediate rivet density is preferable for quasi-static delamination resistance, whereas higher rivet density is required for effective impact damage mitigation. These findings provide useful design guidance for lightweight aerospace composite and sandwich structures subjected to combined static and dynamic loading environments.

References

- [1] W. Wang, *Cohesive zone model for facesheet-core interface delamination in honeycomb FRP sandwich panels*. West Virginia University, 2004.
- [2] X. Wang, Y. An, N. Dou, and Z. Wu, “The peeling strength analysis of sandwich structure with cohesive element model,” in *ITM Web of Conferences*, vol. 47, p. 03016, EDP Sciences, 2022.
- [3] X. Ai, B. Chen, and C. Kassapoglou, “Structural cohesive element for the modelling of delamination in composite laminates without the cohesive zone limit,” *Engineering Fracture Mechanics*, p. 111586, 2025.
- [4] A. Ramaswamy *et al.*, “A discrete cohesive zone model for beam element: Application to adhesively bonded laminates and sandwich panels,” *arXiv preprint arXiv:2410.02432*, 2024.
- [5] G. Yu, X. Chen, B. Zhang, K. Pan, and L. Yang, “Tensile-shear mechanical behaviors of friction stir spot weld and adhesive hybrid joint: Experimental and numerical study,” *Metals*, vol. 10, no. 8, p. 1028, 2020.
- [6] A. Kumar, N. Muthu, and R. G. Narayanan, “Prediction of peel strength of sandwich sheets made of aluminium alloys fabricated by friction stir spot welding based hybrid process using cohesive zone modeling and finite element simulations,” *Engineering Failure Analysis*, vol. 162, p. 108381, 2024.
- [7] S. Maggiore, M. D. Banea, P. Stagnaro, and G. Luciano, “A review of structural adhesive joints in hybrid joining processes,” *Polymers*, vol. 13, no. 22, p. 3961, 2021.
- [8] L. Jiang, B. Yang, S. Xiao, G. Yang, T. Zhu, and D. Dong, “Simulation study of adhesive material for sandwich panel under edgewise compression condition,” *Materials*, vol. 13, no. 6, p. 1391, 2020.
- [9] M. Faidzi, S. Abdullah, M. Abdullah, A. Azman, D. Hui, and S. Singh, “Review of current trends for metal-based sandwich panel: Failure mechanisms and their contribution factors,” *Engineering Failure Analysis*, vol. 123, p. 105302, 2021.
- [10] A. Kumar, P. Saikia, R. G. Narayanan, and N. Muthu, “Experimental and numerical analysis of mixed mode bending of adhesive-bonded and hybrid honeycomb core sandwich structures,” *International Journal of Solids and Structures*, vol. 309, p. 113177, 2025.

- [11] T. McQuigg, R. Kapania, S. Scotti, and S. Walker, “Compression after impact on honeycomb core sandwich panels with thin facesheets, part 1: experiments,” in *53rd AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics and Materials Conference 20th AIAA/ASME/AHS Adaptive Structures Conference 14th AIAA*, p. 1703, 2012.
- [12] P.-S. Shin, J.-H. Kim, Y.-M. Baek, H.-S. Park, D.-J. Kwon, S.-O. Moon, K. L. DeVries, and J.-M. Park, “New evaluation of interfacial and mechanical properties of thermally-treated pine/cfrp composites using electrical resistance measurement,” *Composites Part B: Engineering*, vol. 151, pp. 139–147, 2018.